Master Thesis

A Psychometric Approach for LLM Evaluations

by

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

1.1 General Introduction

DRAFT

LLMs are increasingly used for content generation. However, reliability remains a bottleneck.

Subjective evaluation poses a challenge for Large Language Models (LLMs) and humans alike. Consider an evaluation task, such as the correctness of a mathematical derivation, the number of grammatical errors in a sentence, or the helpfulness of an LLM output. An evaluation task is subjective when the ground truth is not explicitly known, such as for the helpfulness of LLM output: what is helpful to one user might be irrelevant to another. Contrast this with objective evaluation, where output can be validated against an observable ground truth, as is generally the case for e.g. mathematical statements and grammar.

General practice in subjective evaluation is to ask humans to provide 'gold labels'. By training humans to follow a customized evaluation protocol, human evaluators are assumed to approximate a valid measurement of the latent ground truth. However, this process is unscalable.

To automate evaluation, classic Natural Language Generation (NLG) evaluation techniques such as BLEU and ROUGE were often used. Yet, because these only consider surface-level linguistic characteristics like n-gram overlap, they often suffer from low correlation with human evaluations [7]. While useful for simple tasks like translation and summarization quality, they fail to capture the linguistic nuances needed for more complex tasks such as evaluation.

LLMs offer a promising solution. In a seminal paper, Zheng et al. [18] use well-aligned LLMs such as GPT4 in pairwise comparisons, where two candidate texts are compared and the best one is selected. GPT-4 achieves over 80% agreement with human preferences, on par with human-human agreement. Liu et al. [7] adopt a similar approach in their G-Eval framework, prompting LLMs to grade text on a fine-grained Likert scale. Their approach shows significantly higher correlation with human judgements compared to previous state-of-the-art evaluators.

However promising, (limitations of current methods, bridge to subsection 1.2 industrial partner operational challenges)

1.2 Industrial Partner: Barter

The current study is motivated by the operational challenges faced by the industrial partner, Barter. The company aims to leverage Generative AI (GenAI) to streamline the onboarding experience, specifically addressing the friction new partners experience during the sign-up flow. Currently, users must manually write marketing copy such as company descriptions and deal offers before they can make use of the platform. This creates a significant barrier to entry, resulting in user attrition.

While GenAI offers a scalable solution to automate this drafting process, its implementation in business contexts is hindered by the inherent variability of output quality. As LLMs are probabilistic in nature, some degree of inconsistency is unavoidable. Although well-engineered prompts might yield high-quality output in the majority of cases, the accumulation of low-quality output (such as hallucinations and misalignment) becomes a significant operational problem at high volumes.

Two fundamental challenges lie at the core of this problem. First is the challenge of identification: without a reliable mechanism to flag low-quality output, we are unable to quantify the severity of the problem, or make the problematic output actionable. The second is the challenge of diagnosis: improving the output requires an evaluation to identify the reasons why the quality is insufficient. For GenAI to function as an automated solution, it must not only detect the errors but also provide the feedback to resolve them.

Therefore, although Barter's problem statement concerns generation, the solution to the problem begins with accurate validation. Currently, Barter operates in a similar fashion, where employees use GenAI to generate content which is manually evaluated and refined. However, such a human-in-the-loop (HITL) approach is costly and unscalable. To address this, this thesis explores whether small-scale open-source LLMs can be used for automated "AI-in-the-loop" (AITL) evaluation. We introduce a method inspired by the latent variable models of Psychometrics to provide the granular evaluation required for the identification and diagnosis of output quality.

2 Literature Review

2.1 Traditional NLP evaluation methods

2.2 LLM-as-a-Judge

2.3 Measurements through Analytic Decomposition

Prior work provides evidence that analytic decomposition can improve rater reliability or agreement on the decomposed dimensions, but it rarely tests whether aggregating decomposed judgments yields a superior measurement of the corresponding holistic construct compared to a direct holistic evaluation.

3 Theoretical Framework

This thesis treats LLM-based evaluators as measurement instruments that map an input (e.g., marketing copy text) to a score that represents an underlying (subjective) construct (e.g., "Quality"). The goal is to align the LLM evaluation with human evaluation.

The construct of interest can be evaluated directly, in which case it is considered holistically. However, a fundamental challenge in evaluating subjective attributes is that they are not singular, physical properties like temperature or length, but high-dimensional constructs. To address this, we propose a methodology inspired by *construct representation*, treating the task of evaluation as a cognitive process which can be decomposed into sub-processes [9, 16]. We hypothesize that by explicitly modelling the criteria humans implicitly use to reach a judgment, an LLM can yield a better measurement of the construct compared to a holistic evaluation.

A similar measurement approach where a complex construct is decomposed is taken in Li and Sun [4] and Wang et al. [15]. Li and Sun [4] note that it is difficult to align general attributes such as "usefulness" and "harmlessness" with human evaluations, so they decompose them into more specific criteria. Wang et al. [15] find that an "analytic scoring" approach outperforms holistic scoring, where they supply the LLM with Rubrics containing multi-dimensional assessment criteria on which it bases its evaluation.

However, these studies typically treat decomposition as a pragmatic scoring or training design choice, rather than as an explicit construct-specification problem. As a result, it remains unclear how to adequately define the evaluation criteria, how the decomposed scores should be interpreted, and which criteria for validity follow. Validity in psychometrics concerns whether an instrument measures what it is supposed to measure, and serves as support for interpretations and actions based on the measurements. It is established through both theoretical rationales as well as empirical evidence [10]. By making the measurement model explicit, we can define the theoretical requirements for meaningful measurements and derive an empirical validation strategy that matches the construct specification.

Moreover, common choices in the operationalization of LLM-based evaluators can introduce construct-irrelevant variance. In particular, placing multiple criteria in a single prompt makes scores sensitive to criterion order ("position bias"; see Shi et al. [11]), and requesting multiple scores in a single generation mechanically couples later scores to earlier outputs via autoregression. We therefore measure each criterion in isolation, consistent with Bang et al. [1]. Our contribution is to formalize why this operationalization improves measurement validity.

Finally, the validation methodologies in Li and Sun [4] and Wang et al. [15] require improvement. Firstly, they establish *convergent validity* by comparing LLM ratings to human ratings on identical analytic scales. This approach demonstrates that an LLM can mimic human grading on specific sub-dimensions, but fails to prove that the decomposition strategy itself yields a superior measurement of the overall quality compared to a holistic evaluation. The theoretical foundation from Section 3.1 allows us to specify a model to directly assess this. Furthermore, neither paper has access to a dataset which includes behavioral outcomes, limiting their validation to human proxy labels. Demonstrating functional utility is especially important considering the intended use of the evaluator on the Barter platform. We therefore extend validation by assessing *predictive validity* against a downstream behavioral KPI.

Accordingly, the structure of this chapter is as follows.

- **Theoretical definition:** We explain the two causal models that define the relationship between a composite construct ("latent variable") and its constituents ("indicators"): the *reflective* and the *formative* model.
- **Computational operationalization:** We formalize the independent measurement of these constituents by defining the mathematical mapping of dimensions to logit-based scores.
- **Empirical validation:** We explain how we assess measurement validity through the psychometric concept of *construct validity.* We assess construct validity through *convergent validity* (alignment with human holistic labels) and *predictive validity* (prediction of a behavioral KPI).

3.1 Theoretical Definition: Latent Variable Models

Latent variable models define a relationship between a target construct and its indicators. Two model types are distinguished: formative and reflective [2]. They differ in their assumptions about the direction of causality.

In formative models, the latent variable is assumed to be a composite construct, caused by the indicators ("causal indicators"). The canonical example is Socioeconomic Status (SES), which can be defined as the composite of three main features: income, education, and prestige [8]. Formally, for a set of causal indicators indexed by $i = 1, 2, \ldots, n$, we assume

$$\eta = \sum_{i=1}^{n} \gamma_i x_i + \zeta \tag{1}$$

where η is a latent construct, γ_i a formative weight, x_i a manifest variable, ζ the disturbance term.

In reflective models, the latent variable causes the manifest variables ("effect indicators"). For example, following a "g-factor" interpretation of the latent variable intelligence [12], we assume that intelligence is an inherent trait that humans possess which causes people's performance on cognitive tasks. Formally, for any effect indicator i, we assume

$$x_i = \lambda_i \eta + \varepsilon_i \tag{2}$$

where x_i is manifest variable, λ_i a factor loading, η the latent construct, and ε_i the measurement error.

Exemplars in LLM-based measurement. Decomposition already appears in LLM-based evaluation, but with different implicit measurement stances. Wang et al. [15] study LLM scoring in large-scale writing assessment using psychometric reliability frameworks, consistent with treating rubric dimensions as manifestations of a common underlying writing ability (reflective stance). Li and Sun [4] operationalize broad evaluation targets through batteries of criteria organized by task category, emphasizing domain coverage, which is consistent with treating evaluation as a composite defined by its constituents (formative stance).

Implications. These contrasting stances illustrate that the assumed causal model has downstream consequences for both measurement design and validation. Under a reflective specification, indicators are treated as effects of a common underlying attribute; accordingly, strengthening the measurement design (e.g., additional tasks/raters, well-functioning indicators) improves the precision of the latent estimate without redefining the construct, and internal-structure evidence (e.g., factor structure and reliability/generalizability) is a primary diagnostic of measurement quality. Under a formative specification, the construct is defined by its constituents; indicators are not expected to be interchangeable or strongly intercorrelated, and adding or omitting an indicator changes the construct's meaning. Measurement quality therefore hinges on defensible construct specification and content coverage and accurate measurement of each constituent, while validation relies primarily on external relations (e.g., alignment with alternative measures of the target attribute and prediction of relevant criteria).

(relate formative to "content validity", which would be the theoretical requirement for meaningful measurement)

Construct specification (quality-for-purpose). In this thesis, “marketing copy quality” is treated as a quality-for-purpose construct: the attribute is defined by the requirements of its intended use in the Barter setting (i.e., what stakeholders need the copy to achieve). Consequently, the construct is not assumed to be a single underlying trait that causally manifests in multiple observed ratings, but a composite defined by a set of constituent criteria (e.g., clarity, persuasiveness, and compliance with offer details) selected to cover the relevant content domain. This motivates a formative specification: the indicator set constitutes the construct definition, so modifying the set changes what “quality” means. The role of empirical analysis is therefore not to discover a latent factor behind the indicators as in a reflective model, but to test whether the specified composite yields valid measurements with respect to human judgments and downstream behavioral outcomes.

3.2 Operationalization: LLM as a Measurement Instrument

Following the formative specification defined in Section 3.1, we must define a computational process that measures each indicator x_i (corresponding to criterion c_i) independently. We operationalize the LLM as a probabilistic measurement instrument. This section formalizes the mapping from input text to score and demonstrates why standard generation techniques introduce construct-irrelevant variance through positional and autoregressive dependencies.

Input Specification (Encoding) Formally, an LLM maps an input token sequence $\mathbf{u}$ to a distribution of next-token logits $\mathbf{z}(\mathbf{u}) \in \mathbb{R}^{|\mathcal{V}|}$ over a vocabulary $\mathcal{V}$. The conditional probability of a token v is given by the softmax function:

$$P(v \mid \mathbf{u}) = \frac{\exp(z_v(\mathbf{u}))}{\sum_{w \in \mathcal{V}} \exp(z_w(\mathbf{u}))}. \tag{3}$$

In standard evaluation pipelines, multiple criteria are often concatenated into a single input sequence $\mathbf{U}(d) = [\, s\, ;\, c_1\, ;\, \dots\, ;\, c_I\, ;\, d\,]$. However, due to the self-attention mechanism, the representation of any specific criterion c_i becomes a function of its position within $\mathbf{U}$. This introduces position bias, where the measurement is sensitive to the order of instructions rather than the content of the document d [11].

To ensure the measurement of any indicator is invariant to the presence or order of other indicators, we define the input strictly as the triplet of task instruction s, single criterion c_i, and document d. For each criterion i, we construct an independent input sequence $\mathbf{u}_i$:

$$\mathbf{u}_i(d) = [\, s\, ;\, c_i\, ;\, d\,]. \tag{4}$$

This ensures that the logit vector $\mathbf{z}(\mathbf{u}_i)$ depends exclusively on the instructions, the definition of c_i, and the document content.

Score Extraction (Decoding) Standard approaches often require the model to generate a sequence of scores $\mathbf{y} = (y_1, \dots, y_I)$ for the batched criteria. This relies on autoregressive decoding, where the probability of a score y_t is conditioned on the previously generated scores $y_{<t}$:

$$P(\mathbf{y} \mid \mathbf{U}) = \prod_{t=1}^{J} P(y_t \mid \mathbf{U}, y_{<t}). \tag{5}$$

where t denotes the position in the token sequence. This formulation introduces two sources of construct-irrelevant variance. First, it creates *autoregressive coupling*: the measurement of the t-th indicator is mathematically dependent on the values assigned to previous indicators, violating the requirement that a measurement should depend only on the attribute being measured. Second, the generation of a multi-token sequence relies on stochastic sampling strategies (e.g., nucleus sampling) and decoding hyperparameters (e.g., temperature), introducing stochasticity that degrades measurement consistency.

To eliminate these dependencies, we restrict the generation horizon to a single token. We treat the evaluation of criterion c_i as an independent classification task and extract the score y_i by directly analyzing the next-token distribution conditioned on $\mathbf{u}_i(d)$:

$$y_j \sim P(\cdot \mid \mathbf{u}_i(d)). \tag{6}$$

By constraining the "generation" to the first step, we ensure that $P(y_i)$ is conditioned strictly on the input $\mathbf{u}_i(d)$ and never on prior outputs. Furthermore, this approach grants direct access to the logits $\mathbf{z}(\mathbf{u}_i)$, allowing us to bypass the sampling step entirely and derive deterministic, reproducible measurements invariant to decoding strategies.

3.3 Empirical Validation: Construct Validity

In order to determine whether the operationalization aligns with the theoretical constructs, we adopt a *construct validity* approach. Recall that validity is established through both theoretical rationales and empirical evidence [10]. Where Section 3.1 provided the former, this section provides the latter.

The motivation for adopting a construct validity approach is the same for psychology as it is for our evaluations. Unlike objectively measurable properties like "temperature" or "grammatical correctness", the latent (unobservable) nature of constructs means there is no observable "true" value to validate against. Instead, establishing construct validity requires accumulating evidence from multiple sources to demonstrate the appropriateness of the interpretations and their intended use [10]. In this thesis, we build this evidence through two types of construct validity: *Convergent validity* (alignment with human expert judgment) and *predictive validity* (alignment with downstream behavioral outcomes).

Convergent Validity. To assess convergent validity, we treat human expert ratings as a reference measure of the target construct. Let $y^{\mathrm{H}}(d)$ denote a human holistic rating for document d. We assess validity by modeling the relationship

$$y^{\mathrm{H}}(d) = g(\mathbf{x}(d)) + \epsilon(d). \tag{7}$$

This formulation directly implements the formative specification defined in Equation 1 (Section 3.1). By treating the human judgment y^{H} as a proxy for the latent construct η, we can empirically estimate the weights γ that best predict the holistic assessment from the independent indicators.

We evaluate whether the LLM-derived indicators $\mathbf{x}(d)$ explain systematic variation in $y^{\mathrm{H}}(d)$ on held-out data. Importantly, this evaluates the *construct-level* utility of decomposition: rather than only testing whether an LLM can imitate human ratings on the indicator level as in Li and Sun [4] and Wang et al. [15], we test whether the indicator measurements can be combined into a superior predictor of the overall expert judgment.

We compare this against a holistic baseline $x^{\mathrm{hol}}(d)$ obtained by prompting the LLM for a single overall "Quality" score, so we can compare

$$y^{\mathrm{H}}(d) \approx g(\mathbf{x}(d)) \quad \text{vs.} \quad y^{\mathrm{H}}(d) \approx h(x^{\mathrm{hol}}(d)). \tag{8}$$

If formative decomposition improves measurement, the composite model should achieve lower predictive error than the holistic baseline.

Predictive validity. Predictive validity assesses whether a measure exhibits the theoretically expected relationships with an external measure [9]. In our "quality-for-purpose" framework, this corresponds to functional utility: whether the measurements can predict the downstream outcomes the copy is expected to drive.

Let $y^{\mathrm{KPI}}(d)$ denote a behavioral outcome associated with document d (e.g., user engagement or conversion on a platform). We test whether the vector of criterion measurements predicts this outcome:

$$y^{\mathrm{KPI}}(d) = r(\mathbf{x}(d)) + \eta(d), \tag{9}$$

We again compare this performance to the holistic baseline $x^{\mathrm{hol}}(d)$. Evidence that $\mathbf{x}(d)$ is a stronger predictor of $y^{\mathrm{KPI}}(d)$ supports the construct-validity claim that the measurements are appropriate for their intended use, i.e., guiding diagnosis, validation, and improvement of marketing copy in deployment settings.

Together, these two forms of validity provide evidence that our operationalization yields meaningful and usable measurements of the construct under the formative specification. While additional measures of construct validity can be considered (see: Lin [6]), convergent validity and predictive validity target the two principal requirements in our setting: alignment with human evaluations, and functional utility in deployment.

4 Methods

4.1 Data

Below we provide an overview of the two datasets that were used and relevant descriptives. Sample data can be found in Appendix B

4.1.1 FeedbackQA (Human Labels) The *FeedbackQA* dataset Li et al. [5] contains evaluations of Q&A pairs ($n = 5660$). Q&A pairs related to Covid-19 were scraped from the websites of WHO, US Government, UK Government, Canadian government, and the Australian government. Crowd-source workers from English-speaking countries were hired through Amazon's MTurk platform to rate each Q&A pair and provide feedback. The rating is performed on a four-point likert scale (*bad, could be improved, good, excellent*). The feedback concerns "a natural language explanation elaborating on the strengths and/or weaknesses of the answer" [5]. Since each Q&A pair contains two ratings, we aggregate both into the mean to obtain our "ground truth" target variable.

4.1.2 Barter deals (Behavioral KPI) The *Barter Deals* dataset contains 'deals' written by companies ($n = 5111$). On the Barter platform, companies ('partners') can barter goods and services in exchange for influencer content. For example, a company that sells hair styling tools offers their new product in exchange for a video of a creator using it; or, a museum provides free entrance to a creator and their friends in exchange for a video of their visit. The 'deal text' is the marketing copy that is posted on the Barter platform as part of the deals that creators can apply to.

In the current study, we consider the 'deal text' (marketing copy) and the number of applications the deal has obtained after the first seven days of going live (*applications after 7 days*). The variable *applications after 7 days* is chosen because it serves as a behavioral proxy for marketing copy quality. One might apply to a deal because they like the product or the brand, but they could also be convinced to apply bceause of the way the copy is written. A valid measurement of marketing copy quality would therefore be predictive of whether somebody applies to a deal or not. By controlling for the fixed effects of the brand and the product category we can separate the effects of marketing copy quality, and hereby determine whether the LLM measurements satisfy predictive validity.

Table 1. Descriptive statistics of the behavioral KPI *applications after 7 days*

Variable	N	Mean	SD	Min	P25	Median	P75	Max
Applications after 7 days	5111	9.48	16.97	0	0	3	11	188

The KPI is highly dispersed with substantial mass at zero (25th percentile = 0), motivating a count model that accounts for overdispersion.

4.1.3 Text Length Overview To characterize the inputs to the LLM evaluator (and motivate truncation choices), we summarize the distribution of estimated token counts for the two datasets. Table 2 reports standard descriptive statistics.

Table 2. Descriptive statistics of input text length (estimated token counts).

Dataset	N	Mean	SD	Median	Max
FeedbackQA (Q&A text)	5660	166.07	143.83	124	2306
Barter Deals (marketing copy)	5111	79.27	73.65	59	886

The FeedbackQA texts are substantially longer on average than the marketing copy (median: 124 vs. 59 tokens) and exhibit a heavier right tail, with extreme cases exceeding 2000 tokens. This has direct implications for prompt budgeting and truncation. Notably, the marketing copy dataset includes empty texts (minimum token count = 0), which we treat as missing/invalid inputs in downstream analyses.

4.2 Operationalization of Evaluation Constructs

To empirically measure the efficacy of the formative prompting framework, we define a baseline control condition (*holistic prompt*) and an experimental condition (*formative prompt*). In the control condition, we treat the construct holistically and directly measure the construct of interest (*Q&A answer quality* or *marketing copy quality*). In the experimental condition we treat the construct as a formative variable and measure its causal indicators.

4.2.1 Prompt Design & Construct Definition The prompts are specified as the input sequence $\mathbf{u}_i(d) = [\,s\,;\,c_i\,;\,d\,]$, with task instruction s, criterion specification c_i, and document d. We instantiate $\mathbf{u}_i(d)$ in a chat format ("Chat Template") consisting of successive turns with fixed roles: a *system* turn (general instructions), a *user* turn (the document), and an *assistant* turn (the model output). The system message contains s and c_i (in that order), the user message contains d, and generation starts at the assistant turn. We prefix the assistant turn with the literal string `Rating: ` to bias the next-token distribution toward the rating tokens, which was empirically shown to provide more stable measurements. An example chat template is provided in Appendix A.

The criteria c_i provide the LLM with the semantic grounding for the evaluation, and are defined through a qualitative analysis of human evaluations for a particular construct. Recall from Section 3.1 that in a formative specification the construct is defined by the indicators, and omitting an indicator changes the meaning of the construct. Therefore, we select criteria intended to cover the entire construct domain.

FeedbackQA For the question answering task, we derive the formative dimensions from the feedback provided in Li et al. [5]. The authors identify numerous criteria that define "answer quality" by examining the explanations crowd-sourced workers provided for their ratings. From this, we derive three dimensions:

- **Relevance:** The extent to which the answer avoids irrelevant content
- **Completeness:** Whether the answer covers all aspects of the query.
- **Directness:** Whether the answer directly answers the question.

We expect a degree of covariance among the dimensions, since high-quality answers generally score high across all three. However, the formative framework allows for orthogonal variation across the dimensions as well. For example, Li et al. [5] highlight an answer rated *could be improved* (2/4) with the feedback: *"The answer is about some of the online risks but not about how to protect against them."*. This indicates an answer that is relevant but incomplete. While a holistic prompt forces the model to compress the conflicting signals into a single scalar value, the formative prompt explicitly models this through decomposition.

Barter Deals (*this is still in draft, and will be changed*)
For the Barter dataset, we derive the criteria through an analysis of the functional requirements defined by the industrial partner. We reviewed internal guidelines on "effective copy" and identified the key linguistic features hypothesized to drive conversion. The "Marketing Effectiveness" construct is decomposed into:

- **Persuasiveness:** The degree to which the text effectively highlights the value proposition and motivates the reader to consider the offer.
- **Clarity:** The extent to which the offer and terms are communicated unambiguously, minimizing the cognitive effort required to understand the deal.
- **Call-to-Action Strength:** The effectiveness of the final directive in compelling the user to take immediate, concrete action.

To operationalize each criterion c_i as a measurable indicator, we define score-level descriptions for the 1–4 rating scale. By defining each level of the scale, we provide additional context that anchors the intended interpretation of the ratings and reduces ambiguity in how the model applies the scale.

4.3 Model Selection & Implementation

To perform evaluations we select three popular small-scale, open-source instruction-tuned models: Qwen3-4B-Instruct-2507 [14] (AliBaba), Gemma3-4B-IT [13] (Google), and Llama-3.2-3B-Instruct (Meta). These models were selected to represent the current state-of-the-art open-source models. Notably, Qwen has previously shown to outperform other similar sized open-source LLMs in evaluation tasks [3].

The choice of model type (small, open-source, instruction-tuned) is based on the methodological requirements and the operational constraints of the industrial partner. Methodologically, the use of open-source models enables "white-box" access to the model's probability distribution. This allows us to compute the logit-weighted mean unavailable in standard black-box APIs. Moreover, we prefer instruction-tuned models over base models to ensure adherence to the evaluation instructions provided in the prompt. Operationally, Barter requires the evaluator to balance measurement validity with cost-efficiency and latency. Unlike proprietary models (e.g., GPT-4) where costs scale linearly with volume, small open-source models incur only fixed compute costs, making them a more scalable solution for high-volume deployment.

We implement the inference pipeline using the HuggingFace `transformers` library [17]. To facilitate the single-token scoring method, we developed a custom generation wrapper that bypasses the standard autoregressive loop. Instead of sampling text, this wrapper performs a single forward pass on the prompt sequence and extracts the raw unnormalized logits.

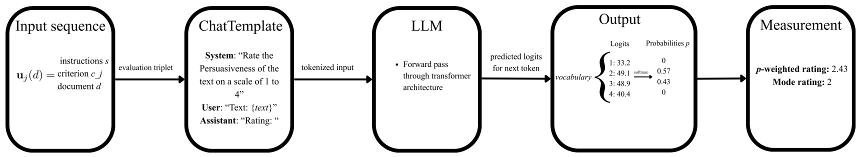

Fig. 1. Logit-Based LLM Evaluation Pipeline. The input sequence is processed through a frozen transformer architecture (single forward pass) to generate logits for the next token. Rather than sampling text, the evaluator extracts probabilities for specific rating tokens (1–4) to obtain evaluations. N.b.: Specifications are ommitted from the ChatTemplate for brevity, for a full example see Appendix A.

4.4 Computation of evaluations

The evaluation triplet $\mathbf{u}_i(d)$ is supplied in "Chat Template" format to the LLM, and we perform a single forward pass to obtain the next-token logits $\mathbf{z}(\mathbf{u}) \in \mathbb{R}^{|\mathcal{V}|}$. We subset the logits for the tokens $\mathcal{S} = \{`1', `2', `3', `4'\}$, adhering to the four-point Likert scale in Li et al. [5]

To obtain the evaluations, we compute the mode and the weighted mean of the constrained logit distribution. Let $v(k)$ be a function that maps a token string k to its integer value. The *modal score* x_i^{mode} corresponds to a standard greedy decoding procedure (temperature $T = 0$) and is computed as:

$$x_i^{\text{mode}} = v\left(\underset{k \in \mathcal{S}}{\text{argmax}}\; P_{\mathcal{S}}(k \mid \mathbf{u}_i)\right). \tag{10}$$

The *expected score* x_i^{mean} is computed by first normalizing the logits over $\mathcal{S}$ via a softmax function to obtain probabilities, and then calculating the probability-weighted sum:

$$x_i^{\text{mean}} = \sum_{k \in \mathcal{S}} v(k) \cdot P_{\mathcal{S}}(k|\mathbf{u}_i) \tag{11}$$

By considering the token probabilities we incorporate model estimates of rating uncertainty, instead of collapsing this information into an integer score through the mode. As Liu et al. [7] show, this more fine-grained approach can provide better evaluations.

4.5 Experimental design & Validation Strategy

4.5.1 Alignment with Human Raters (FeedbackQA) The target variable is the average of the two human evaluations of Q&A answer quality, noting that the choice between averaging or selecting one rating column over the other yields no qualitative differences in the results.

Correlation We use Spearman ρ to compute the monotonic (rank) correlation between the human evaluations and the LLM evaluations, which is interpreted as a general indicator for criterion relevance. Moreover, we compute the correlation between the formative dimensions as a diagnostic for whether the LLMs adequately distinguish the different criteria.

Convergent validity. To assess convergent validity, we test whether LLM measurements predict human evaluations of *Q&A answer quality.*

Following the formative specification in Equation 7, we instantiate the construct-level mapping $g(\mathbf{x}_d)$ as a linear model and estimate:

$$y_d^{\mathrm{H}} = \beta_0 + \beta_1 x_{\mathrm{rel},d} + \beta_2 x_{\mathrm{comp},d} + \beta_3 x_{\mathrm{dir},d} + \varepsilon_d, \tag{12}$$

for Q&A pair $d = 1, \ldots, N$. Here, y_d^{H} denotes the human holistic rating of answer quality for pair d, and $x_{\mathrm{rel},d}$, $x_{\mathrm{comp},d}$, and $x_{\mathrm{dir},d}$ denote the LLM measurements of relevance, completeness, and directness, respectively. The β coefficients correspond to the formative weights γ from Equation 1.

4.5.2 Behavioral Prediction (Barter Deals) We assess *predictive validity* by testing whether the LLM-derived formative indicators predict the behavioral KPI *applications after 7 days.* The underlying hypothesis is that higher-quality marketing copy increases user engagement, which should translate into more applications within the first week. We specify a time-period of seven days since deals can be left online for unspecified amounts of time, garnering more applications. One week serves as a consistent measure of deal engagement, accounting for possible time-of-day or weekday effects.

The KPI is a non-negative count variable and is strongly right-skewed. Accordingly, we model `applications_till_day_7` as a Generalized Linear Model (GLM) for count data. Let y_d denote the number of applications within 7 days for deal d, and let $\hat{x}_i(d)$ be the LLM-measured score for indicator i. We specify a Negative Binomial GLM with log link:

$$y_d \mid \mu_d \sim \mathrm{NegBin}(\mu_d, \kappa), \tag{13}$$

$$\log(\mu_d) = \beta_0 + \sum_{i=1}^{I} \beta_i \hat{x}_i(d) + \gamma^\top \mathrm{category}_d, \tag{14}$$

where $\mathrm{category}_d$ denotes category fixed effects, included to control for systematic differences in baseline application volume across categories.

5 Results

5.1 Convergent Validity (FeedbackQA)

(note that I still need to generate results on the full data sample, but the results will remain approximately the same)

We first evaluate the extent to which the formative operationalization aligns with human expert judgments compared to the holistic baseline. Table 3 presents the Adjusted Coefficient of Determination (R^2_{adj}) for both specifications across all three target models.

Table 3. Comparative performance of Holistic vs. Formative specifications in predicting human expert ratings on FeedbackQA. The Formative model consistently explains greater variance in the ground truth.

Model	**Holistic Baseline** (R^2_{adj})	**Formative Model** (R^2_{adj})	**Improvement** (Δ)
Qwen 3-4B	0.387	**0.432**	+0.045
Llama 3.2-3B	0.318	0.419	+0.101
Gemma 3-4B	0.263	0.302	+0.039

The results provide strong empirical support for the formative hypothesis. Across all evaluated models, the formative specification yields a higher R^2_{adj} than the holistic baseline, indicating that the decomposed indicators captures systematic variation in quality that the single-token holistic score misses.

Model-Specific Performance. Qwen3-4B achieves the highest overall alignment with human raters ($R^2_{adj} = 0.432$), confirming its status as the strongest model in this size class. However, Llama 3.2-3B demonstrates the most significant benefit from decomposition. While its holistic performance is moderate (0.318), the formative specification increases predictive to 0.419, effectively closing the gap with Qwen.

This disparity suggests that while Llama possesses the latent capability to identify quality features (as evidenced by the high formative score), it struggles to aggregate these signals into a single scalar during inference. The formative operationalization effectively bypasses this "aggregation bottleneck."

5.2 Barter Deals

6 Discussion

Bibliography

[1] G. Bang, D. Kim, and M. Min. Transforming user defined criteria into explainable indicators with an integrated LLM AHP system. URL http://arxiv.org/abs/2601.05267.

[2] K. Bollen and R. Lennox. Conventional wisdom on measurement: A structural equation perspective. *Psychological bulletin*, 110(2):305, 1991.

[3] J. Gu, X. Jiang, Z. Shi, H. Tan, X. Zhai, C. Xu, W. Li, Y. Shen, S. Ma, H. Liu, S. Wang, K. Zhang, Y. Wang, W. Gao, L. Ni, and J. Guo. A survey on LLM-as-a-judge. URL http://arxiv.org/abs/2411.15594. Read_Status: In Progress Read_Status_Date: 2025-10-14T13:58:05.029Z.

[4] Y. Li and Y. Sun. A fine-grained evaluation framework for language models: Combining pointwise grading and pairwise comparison. 63(1): 104270. ISSN 0306-4573. https://doi.org/10.1016/j.ipm.2025.104270. URL https://www.sciencedirect.com/science/article/pii/S0306457325002110.

[5] Z. Li, P. Sharma, X. H. Lu, J. C. K. Cheung, and S. Reddy. Using interactive feedback to improve the accuracy and explainability of question answering systems post-deployment. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 926–937. https://doi.org/10.18653/v1/2022.findings-acl.75. URL http://arxiv.org/abs/2204.03025.

[6] Z. Lin. A validity-guided workflow for robust large language model research in psychology. URL http://arxiv.org/abs/2507.04491. Read_Status: In Progress Read_Status_Date: 2025-10-14T14:31:31.251Z.

[7] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu. G-eval: NLG evaluation using GPT-4 with better human alignment. URL http://arxiv.org/abs/2303.16634.

[8] S. McQuitty and M. Wolf. Structural equation modeling: A practical introduction. *Journal of African Business*, 14(1):58–69, 2013.

[9] S. Messick. Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. 1994(2):i–28. ISSN 2330-8516. https://doi.org/10.1002/j.2333-8504.1994.tb01618.x. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/j.2333-8504.1994.tb01618.x. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/j.2333-8504.1994.tb01618.x Read_Status: New Read_Status_Date: 2025-10-14T12:42:52.831Z.

[10] S. Messick. Meaning and values in test validation: The science and ethics of assessment. *Educational researcher*, 18(2):5–11, 1989.

[11] L. Shi, C. Ma, W. Liang, X. Diao, W. Ma, and S. Vosoughi. Judging the judges: A systematic study of position bias in LLM-as-a-judge. URL http://arxiv.org/abs/2406.07791.

[12] C. Spearman. " general intelligence" objectively determined and measured. 1961.

[13] G. Team. Gemma 3. 2025. URL https://goo.gle/Gemma3Report.
[14] Q. Team. Qwen3 technical report, 2025. URL https://arxiv.org/abs/2505.09388.
[15] Y. Wang, J. Huang, L. Du, Y. Guo, Y. Liu, and R. Wang. Evaluating large language models as raters in large-scale writing assessments: A psychometric framework for reliability and validity. *Computers and Education: Artificial Intelligence*, page 100481, 2025.
[16] S. E. Whitely. Construct validity: Construct representation versus nomothetic span. *Psychological bulletin*, 93(1):179, 1983.
[17] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, et al. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 conference on empirical methods in natural language processing: system demonstrations*, pages 38–45, 2020.
[18] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. URL https://www.semanticscholar.org/paper/Judging-LLM-as-a-judge-with-MT-Bench-and-Chatbot-Zheng-Chiang/a0a79dad89857a96f8f71b14238e5237cbfc4787.

A Prompt rendering via `apply_chat_template`

We format each prompt using Hugging Face's chat template with `add_generation_prompt=True`, and then append an explicit assistant prefix (shown below). The rendered template is passed to the model, which computes the next-token probabilities after the assistant prefix. The model input is the concatenation of:

(i) system message, (ii) user message, (iii) assistant header inserted by the template, (iv) `assistant_prefix`.

```
<System role header as produced by tokenizer>
Instruction: Rate the Relevance of the Answer provided for
    the Question
on a scale of 1 to 4. Use the following rubric to determine
    the score:

Relevance: Does the answer discuss the specific topic and
    entities requested?

1 = Irrelevant: Discusses a completely different topic (e.g.,
     "baby delivery"
instead of "respirators").

2 = Topic Mismatch: Discusses a related category but the
    wrong specific
entity (e.g., "Work Visa" instead of "Pandemic Visa").

3 = Broadly Relevant: Discusses the correct topic but
    includes significant
tangential or unrelated information.

4 = Precise: Focuses exclusively on the specific entity and
    topic requested
in the question.
</System role header>

<User role header as produced by tokenizer>
Question:
{question}

Answer:
{answer}
</User role header>

<Assistant role header inserted by add_generation_prompt=True
    >
Rating:
```

B Sample data

B.1 FeedbackQA examples

Example F01

Question: Is it still possible to access medical appointments and medications through my doctor?

Answer: You can access bulk-billed appointments with your doctor, nurse or mental health professional via phone or videoconference. If you would like to arrange a telehealth appointment, contact your doctor. Accessing medicines while staying at home. Your doctor may be able to arrange to have your prescriptions sent to your pharmacy directly so you can stay at home. For more information on whether you can access this service please call your pharmacy and speak to them about the best way to arrange this.

Rating 1: 4

Feedback 1: Gives information on how patients can access appointments with their doctors during the pandemic.

Rating 2: 3

Feedback 2: This answer relates to accessing medical appointments and medications, but does not give much information.

Table 4. Sampled FeedbackQA items (IDs and labels), and Qwen scores for Relevance, Completeness, Directness, and Holistic criteria

ID	Human label 1	Human label 2	Rel.	Comp.	Dir.	Holistic
F01	4	3	4	3	3	4

B.2 Barter Deals examples

Example B01

Deal text: We'd love a video capturing the experience of visiting our store in the heart of Amsterdam—from the vibe of the city and the walk over, to exploring the shop itself. While you're there, pick your favorite bag from the collection and make it part of the story!
Keep it authentic, stylish, and feel free to show the charm of the surroundings too!

Applications after 7 days: 39

Product category: Activities

Example B02

Deal text: Choose 2 coffees and pastries you like:)

Applications after 7 days: 6

Product category: Food